

METplus: A one-stop shop for trusted, reproducible, and scalable verification capabilities

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6 November 2025 | *harrold@ucar.edu

METplus: A one-stop shop for trusted verification

METplus is...

- A highly configurable verification framework that spans a range of temporal and spatial scales
- Community-driven, extensible, scalable

METplus has...

- Over 150 statistics and diagnostics for point and gridded datasets
- User support via GitHub Discussions
- Use cases for easy spin-up

METplus provides...

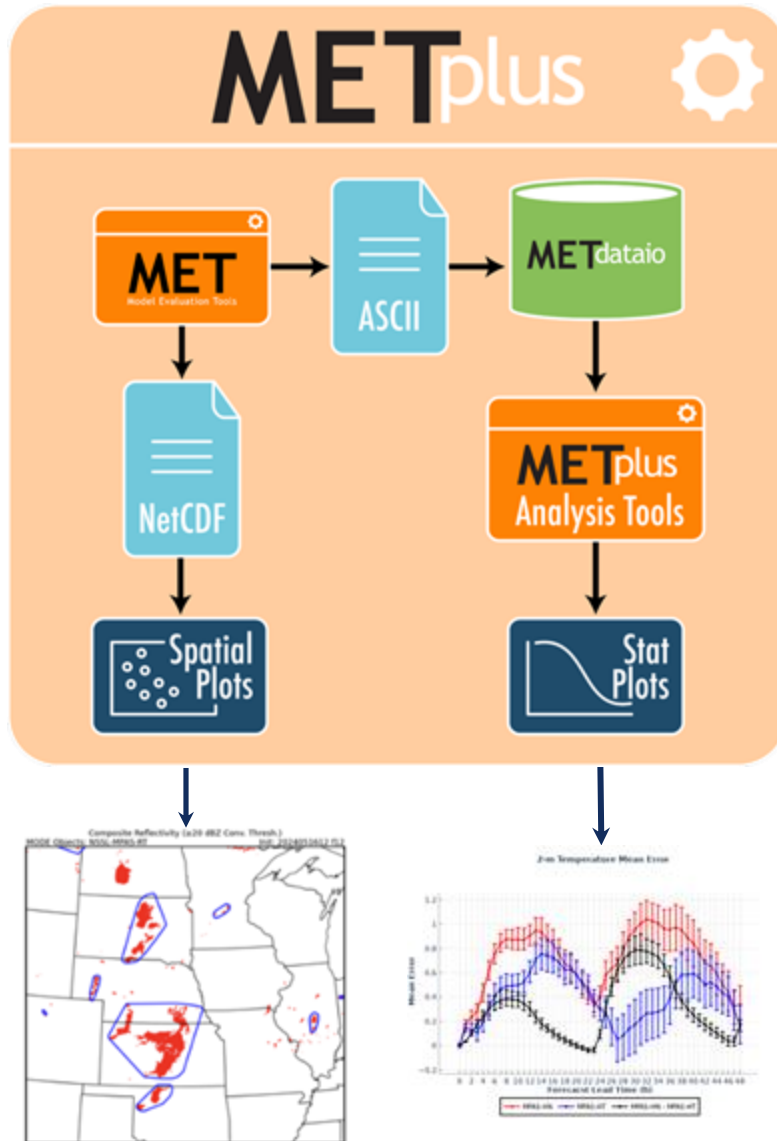
- A versatile toolkit that enables objective and reproducible evaluations over a broad range of weather and climate applications

METplus is open source and community driven with active partnerships in:

- ✓ National and international operational centers
- ✓ Universities
- ✓ Research community
- ✓ Private sector



METplus: A one-stop shop for trusted verification



METplus ecosystem components:

- **Wrappers** – low level workflow (black arrows)
- **MET** – suite of statistical and diagnostic tools
- **Analysis Tools** – available for advanced analysis, includes:
 - METviewer and METexpress user interface
 - METdataio, METcalcpy, METplotpy components

METplus verification capabilities:

- Comprehensive point-based and grid-based traditional verification and spatial verification
- Deterministic, ensemble, and probabilistic verification
- Extensive capabilities for auto-regridding, interpolating, masking, reformatting data...and more!

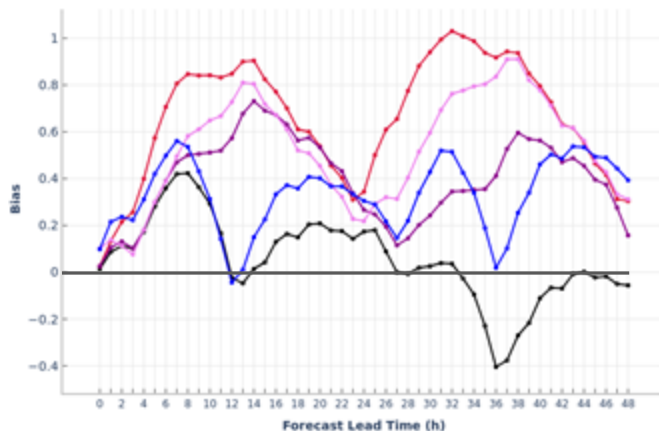
Examples of statistics & diagnostics

Examples of traditional verification:

- Grid-to-grid and grid-to-point
- Deterministic, ensemble, and probabilistic
- Single initialization and time-lagging

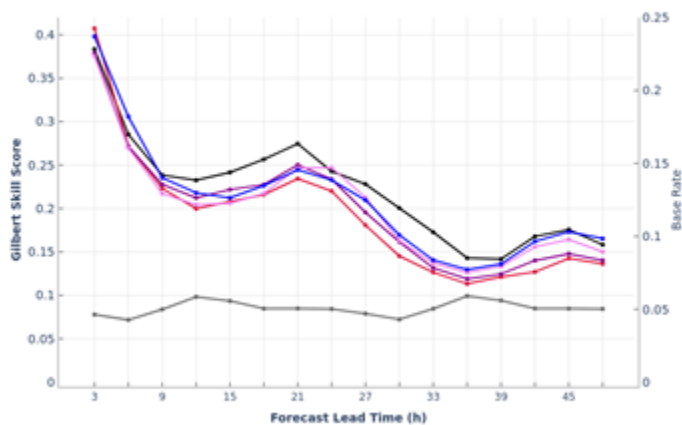
Graphics generated with METplus Analysis Tools and user-generated Python scripts.

2-m Temperature Bias (K): 29 April - 31 May 2024



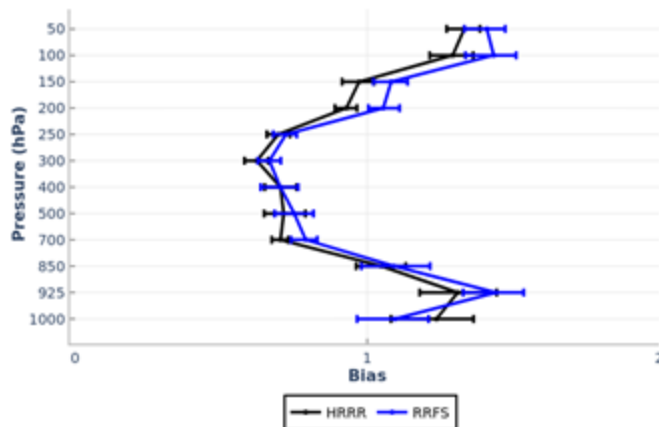
— HRRR — NSSL-MPAS-HN — NSSL-MPAS-HT — NSSL-MPAS-RT — RRFS

3-h Accum. Precip. (≥ 0.1 in) GSS: 29 April - 31 May 2024



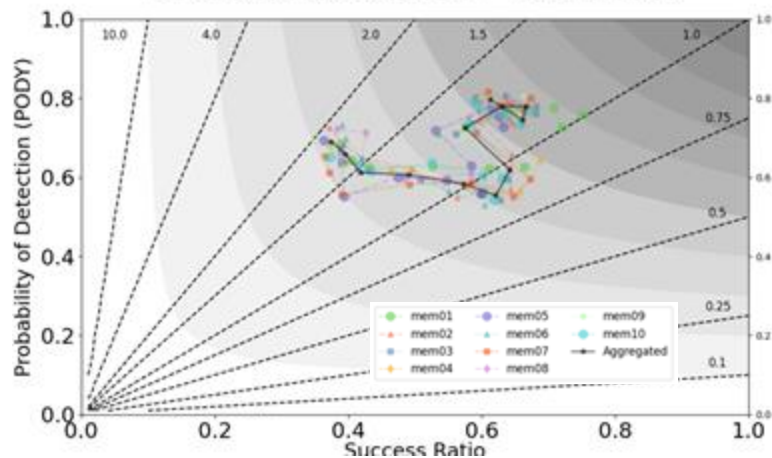
— HRRR — NSSL-MPAS-HN — NSSL-MPAS-HT — NSSL-MPAS-RT — RRFS — Base Rate

Temperature BCRMSE (K): July 2023 (00 UTC Inits; 12 h Lead Times)



— HRRR — RRFS

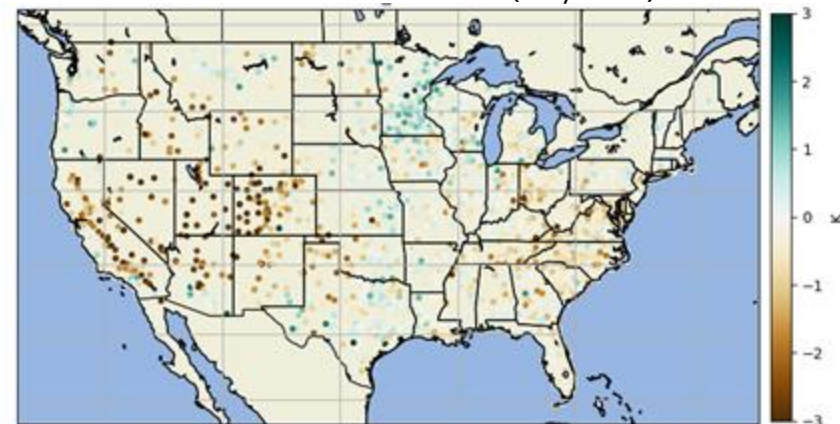
Performance Diagram (APCP3h - May 3rd, 2022)



Scorecard: HRRR and RRFS

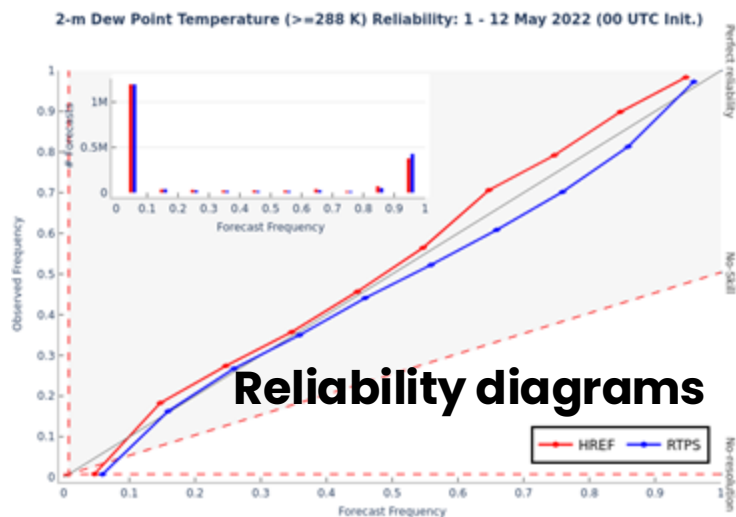
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HRRR: 2-m Dew Point Bias (May 2024)



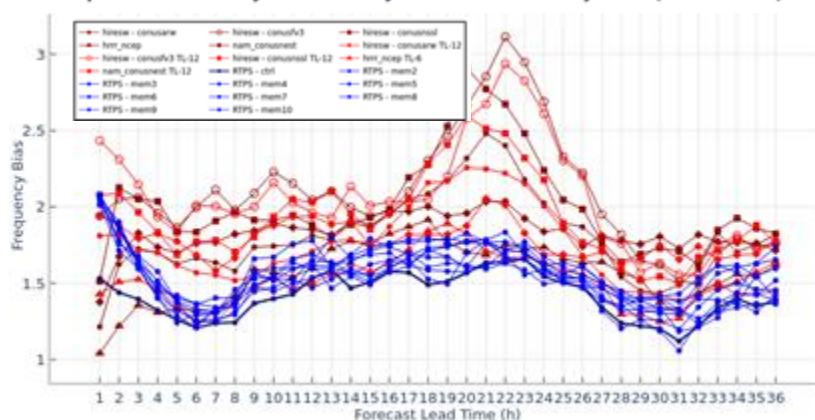
Examples of traditional verification:

- Grid-to-grid and grid-to-point
- Deterministic, ensemble, and probabilistic
- Single initialization and time-lagging

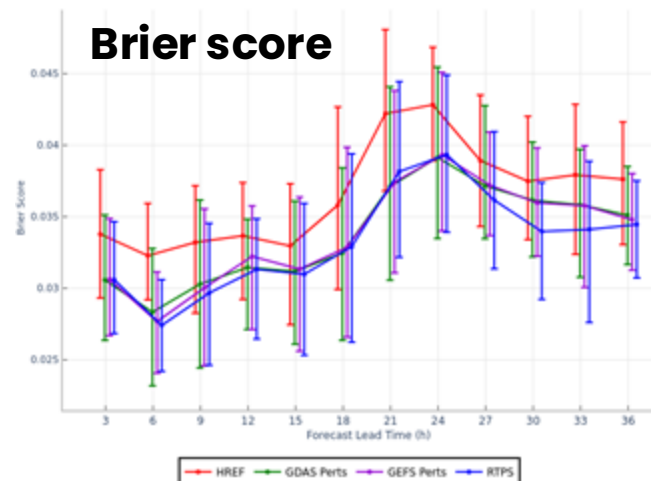


Single initialization and time-lagging

Composite Reflectivity ≥ 20 dBZ by Member: 1 - 12 May 2022 (00 UTC Init.)

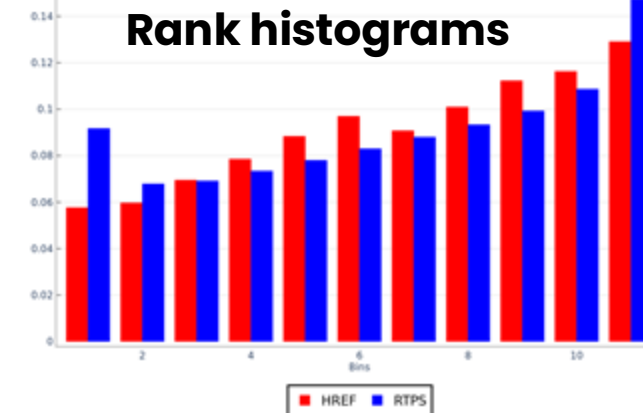


Composite Reflectivity (≥ 20 dBZ) Brier Score: 1-6 May 2022 (00 UTC Init.)

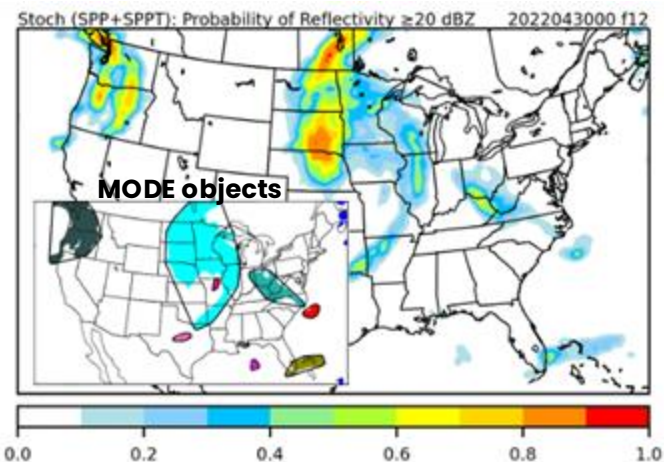


2-m Temperature Rank Histogram: 1-12 May 2022 (00 UTC Init.)

Rank histograms

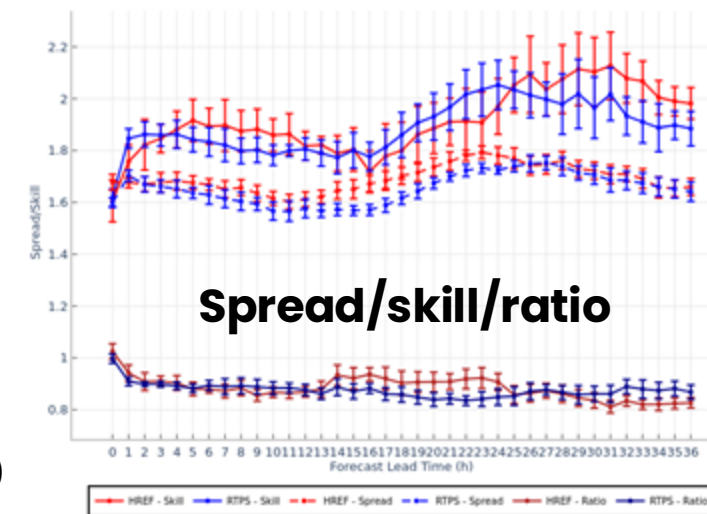


Probabilistic verification



Neighborhood Ensemble Probabilities (NEP)
of a 10 member RREFS-based ensemble

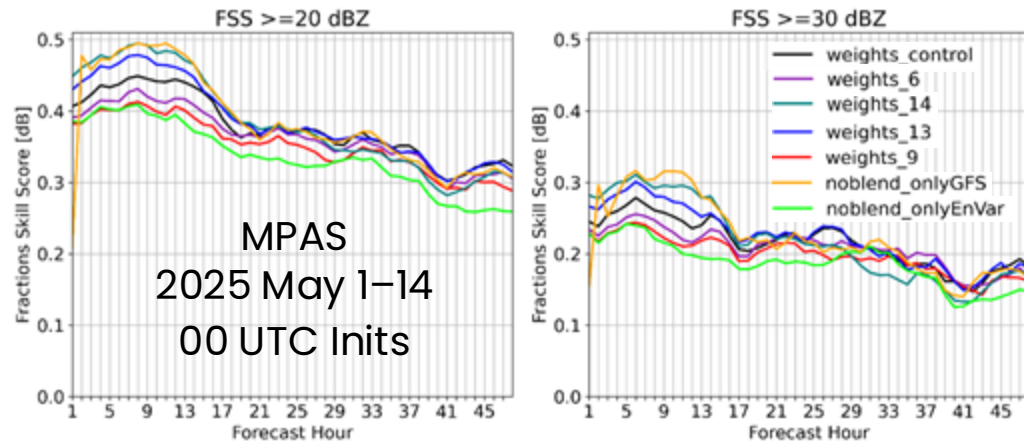
10-m Wind Speed (m/s): 1 - 12 May 2022 (00 UTC Init.)



Examples of spatial verification:

- Neighborhood methods (e.g., FSS)
- Object-based verification with MODE

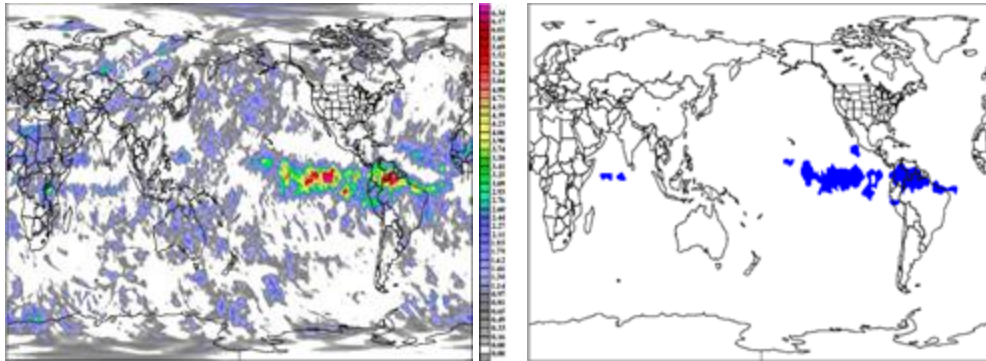
Example of FSS



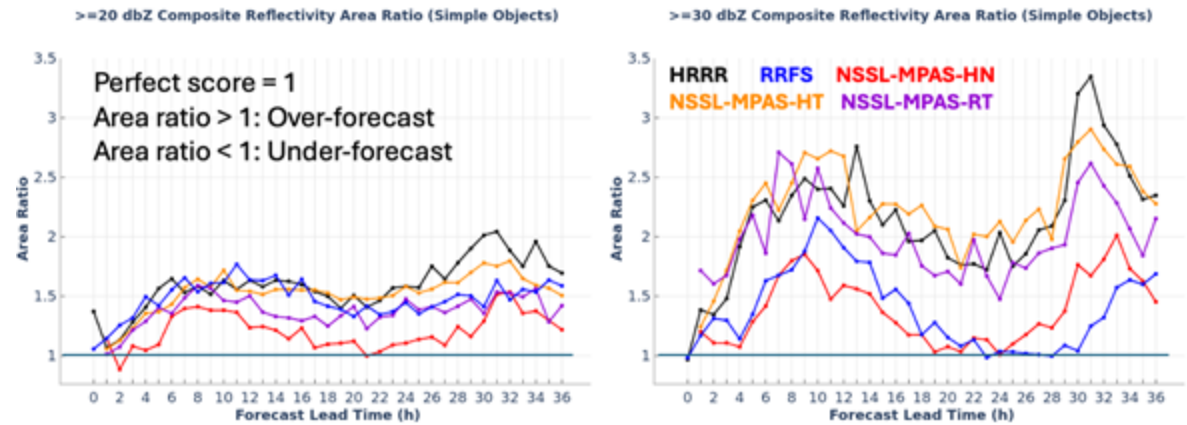
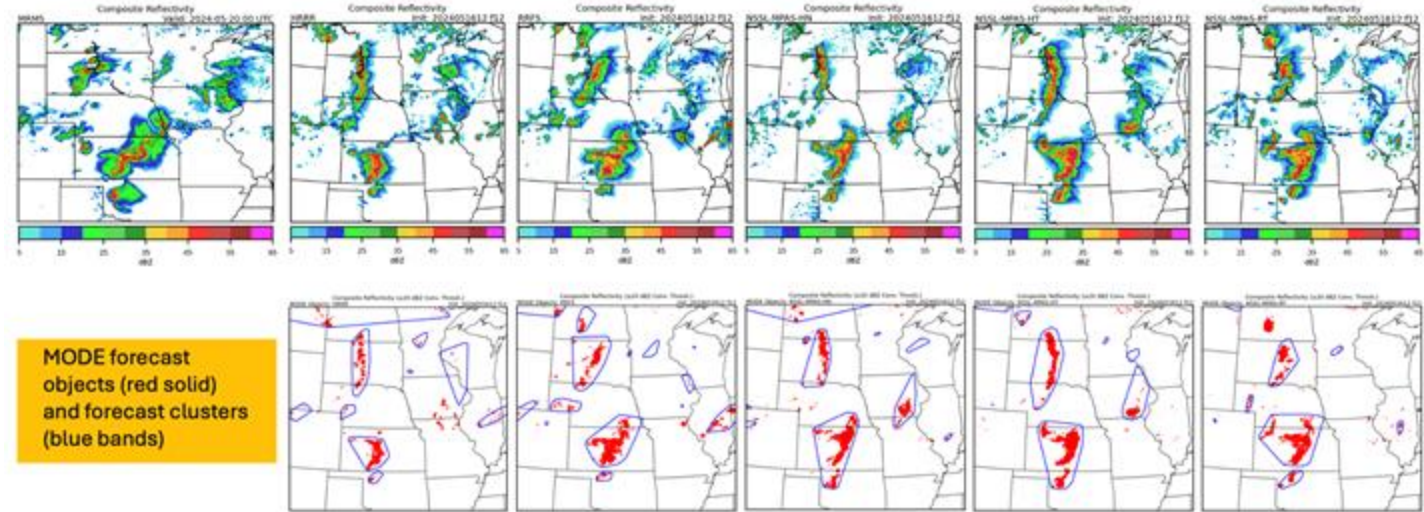
Example of MODE for validation

UWIND forecast differences

UWIND objects



Example of MODE on 2024 HWT-SFE data



Examples of tropical cyclone verification:

- Standard track and intensity
- Radius of maximum wind diagnostics
- Object-based evaluation

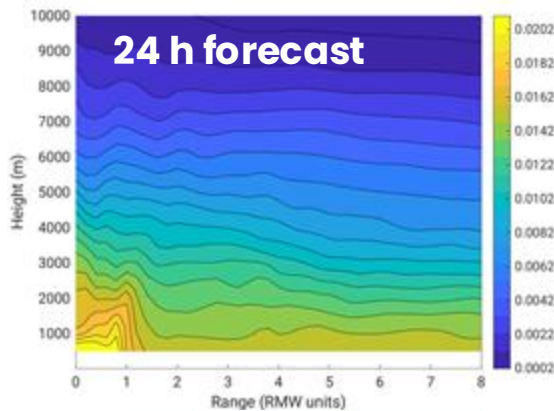
Radius of maximum wind (RMW) diagnostics (Hurricane Idalia)

6-h Accum. Precip.
Storm Relative,
Normalized by RMW

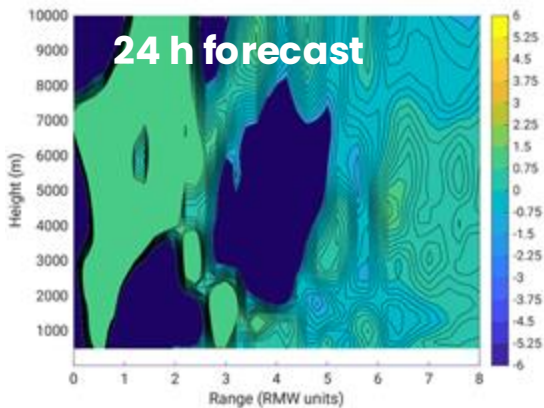
OBS

Forecast

Azimuthally averaged
specific humidity



Azimuthally averaged
vertical velocity

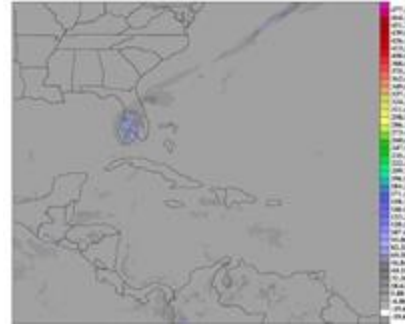


Object-based verification with MODE (Hurricane Ian)

Forecast Objects

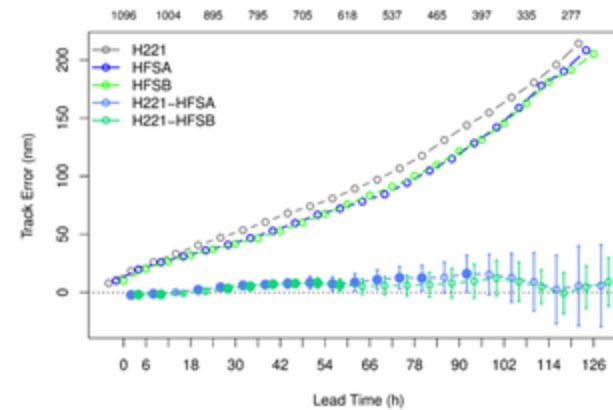
Observed Objects

Forecast Objects w/
Observed Object Outlines

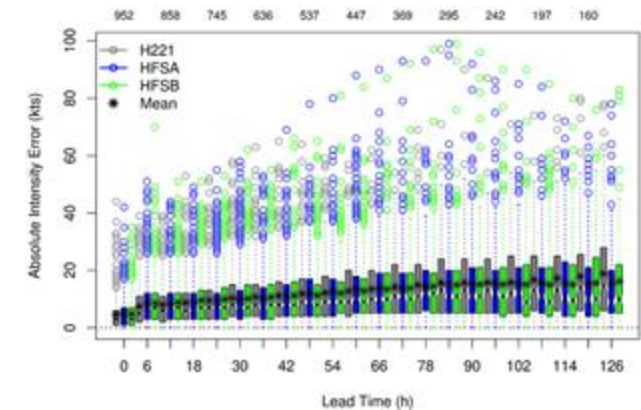


Track and intensity verification

Mean Track Error
Atlantic Basin



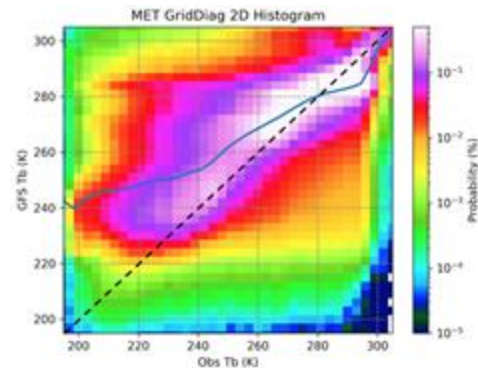
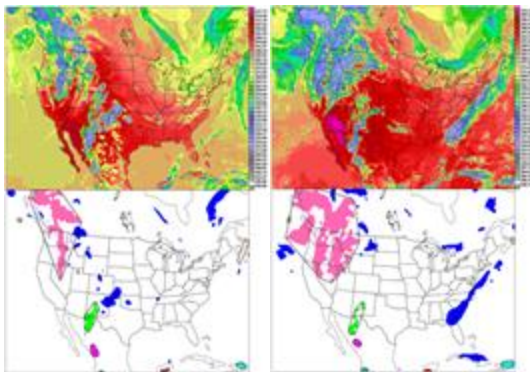
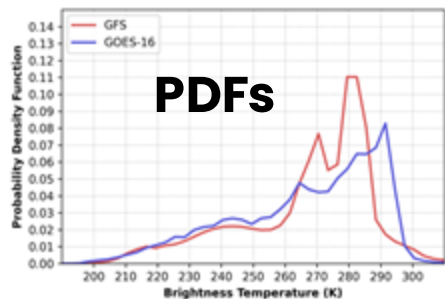
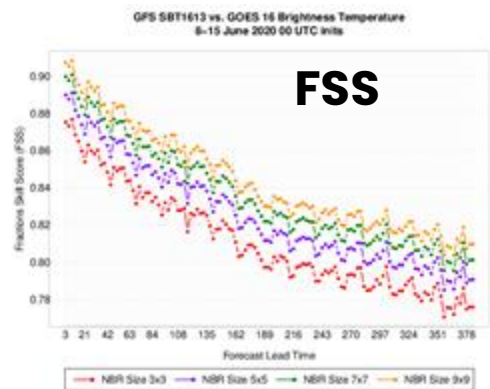
Mean Absolute Intensity Error
Eastern Pacific Basin



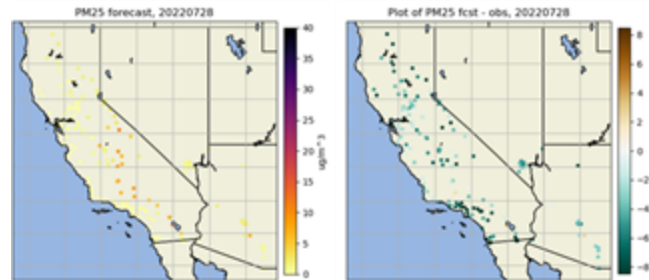
Examples of application verification:

- Clouds
- Dust/smoke/aerosol
- Seasonal-to-subseasonal
- Space weather
- Fire weather verification

Cloud verification

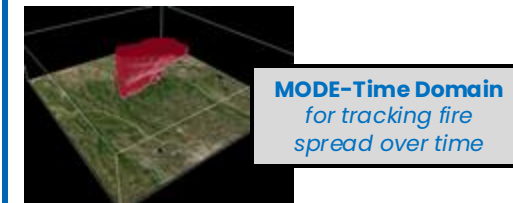


Dust/smoke verification

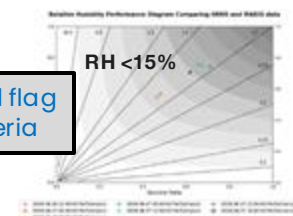


✓ Particulate matter, aerosol optical depth, smoke, and visibility

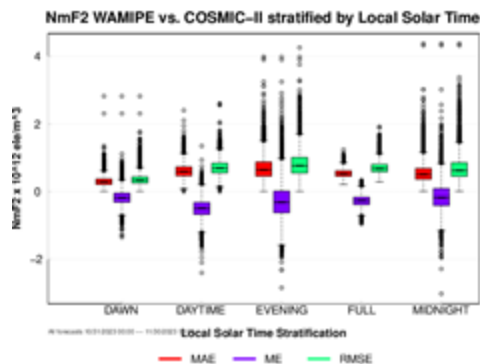
Fire verification



Evaluating red flag warning criteria

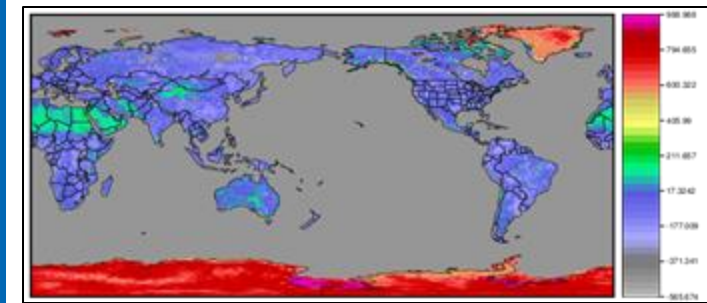


Solar verification



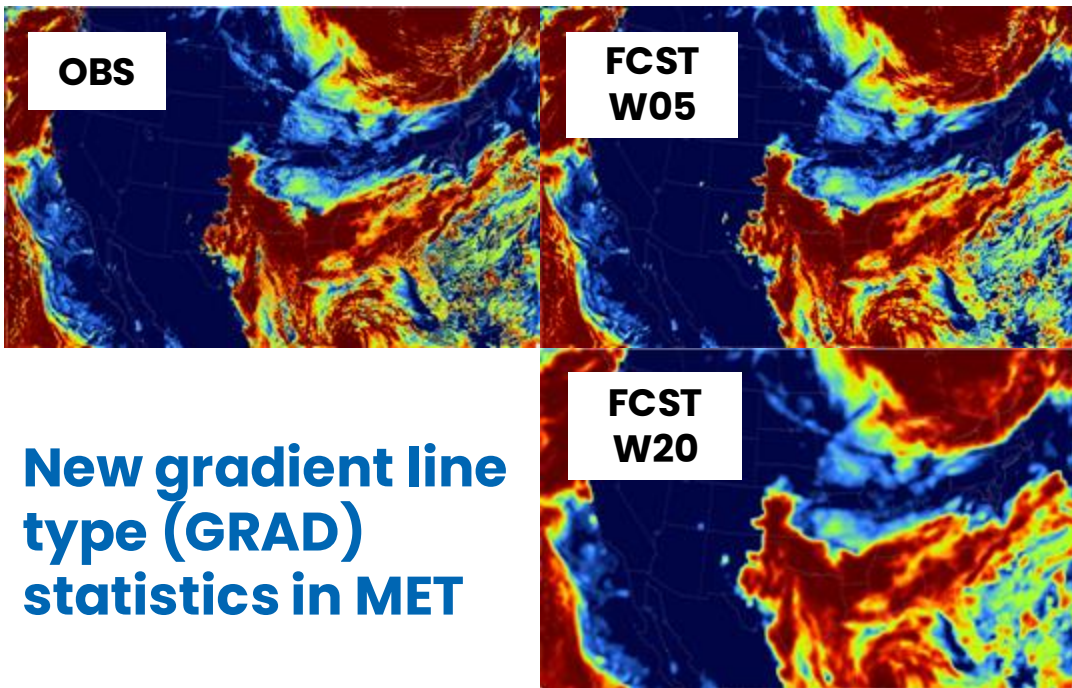
S2S verification

SFS-GSL Model
Soil Moisture Mean Error
(June, 1991-2020)



Examples of AIWP verification:

- Gradient statistics in MET
- Examples of verifying CREDIT



New gradient line type (GRAD) statistics in MET

	FGMAG	OGMAG	MAG_RMSE	LAPLACE_RMSE
W05	1.92	4.25	8.74	11.13
W20	0.82	4.25	10.36	12.27

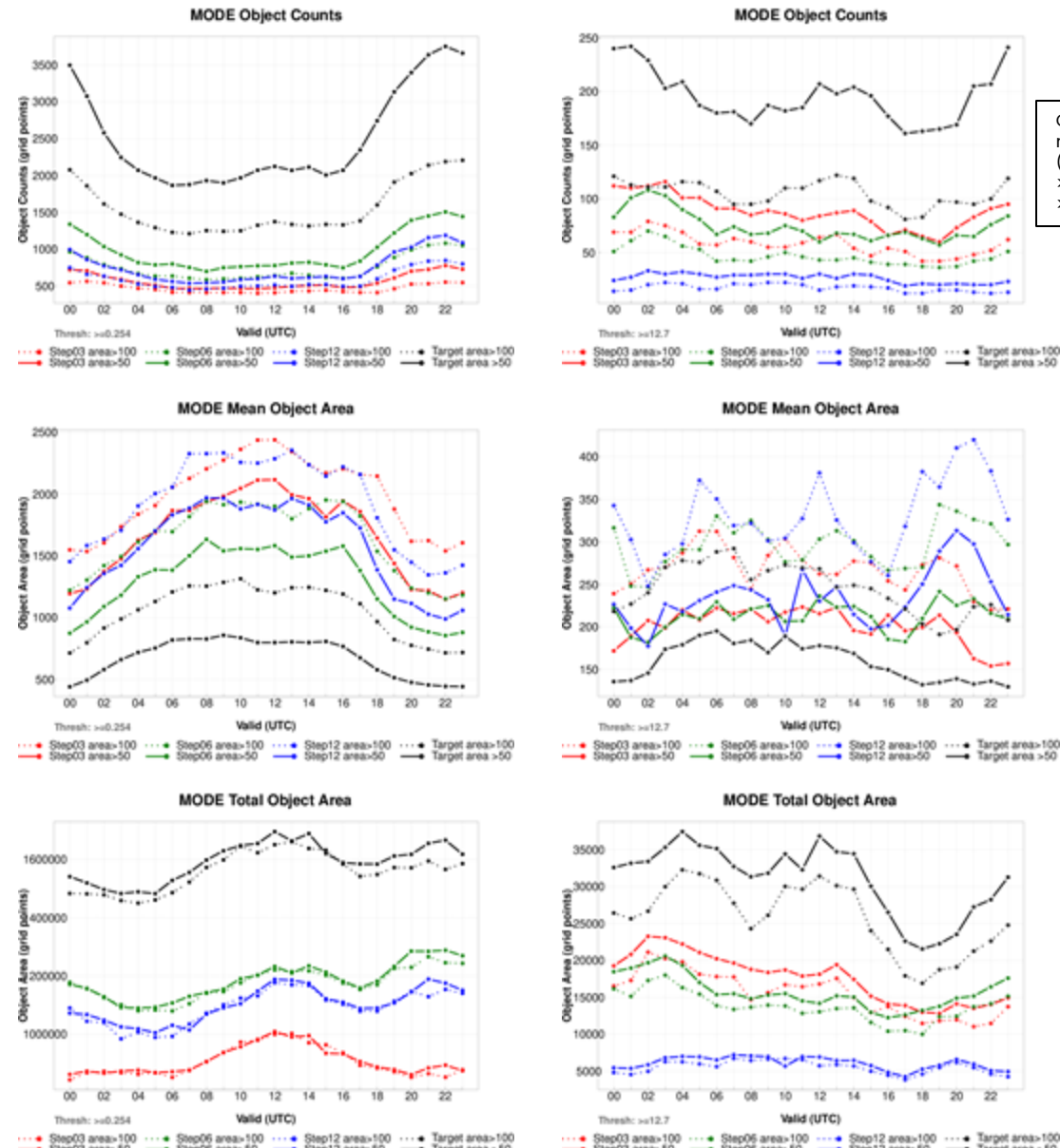
Sharpness metrics (Ebert-Uphoff et al. 2025)

Object-based verification

1-h Accumulated Precipitation

Threshold: ≥ 0.01 in

Threshold: ≥ 0.5 in



METplus aiding in reproducible science



METplus configuration files

What are they?

- METplus configuration files that define:
 - ✓ Processes to run – MET tools and user scripts
 - ✓ Run times – initialization/valid and forecast leads
 - ✓ Paths using filename templates to find files for each time
 - ✓ MET configuration settings
 - ✓ Environment variables

```
[config]
# Documentation for this use case can be found at
# https://metplus.readthedocs.io/en/latest/generated/met_tool_wrapper/RegridDataPlane/RegridDataPlane.html
# For additional information, please see the METplus Users Guide.
# https://metplus.readthedocs.io/en/latest/Users_Guide

###
# Processes to run
# https://metplus.readthedocs.io/en/latest/Users_Guide/systemconfiguration.html#process-list
###

PROCESS_LIST = RegridDataPlane

###
# Time Info
# LOOP_BY options are INIT, VALID, RETRO, and REALTIME
# If set to INIT or RETRO:
#   INIT_TIME_FMT, INIT_BEG, INIT_END, and INIT_INCREMENT must also be set
# If set to VALID or REALTIME:
#   VALID_TIME_FMT, VALID_BEG, VALID_END, and VALID_INCREMENT must also be set
# LEAD_SEQ is the list of forecast leads to process
# https://metplus.readthedocs.io/en/latest/Users_Guide/systemconfiguration.html#timing-control
###

LOOP_BY = INIT
INIT_TIME_FMT = %Y%m%d%H
INIT_BEG=2005080700
INIT_END=2005080700
INIT_INCREMENT = 1M

LEAD_SEQ = 3H

###
# File I/O
# https://metplus.readthedocs.io/en/latest/Users_Guide/systemconfiguration.html#directory-and-
###

OBS_REGRID_DATA_PLANE_RUN = True

OBS_REGRID_DATA_PLANE_INPUT_DIR = {INPUT_BASE}/met_test/data/sample_obs/ST2m1
OBS_REGRID_DATA_PLANE_OUTPUT_DIR = {OUTPUT_BASE}/regrid_data_plane
OBS_REGRID_DATA_PLANE_TEMPLATE = ST2m1(valid?fmt=%Y%m%d%H).Grb_G212

###
# Field Info
# https://metplus.readthedocs.io/en/latest/Users_Guide/systemconfiguration.html#field-info
###

OBS_REGRID_DATA_PLANE_VAR1_INPUT_FIELD_NAME = APCP
OBS_REGRID_DATA_PLANE_VAR1_INPUT_LEVEL = A01
OBS_REGRID_DATA_PLANE_VAR1_OUTPUT_FIELD_NAME = APCP_01

REGRID_DATA_PLANE_ONCE_PER_FIELD = True

###
# RegridDataPlane Settings
# https://metplus.readthedocs.io/en/latest/Users_Guide/wrappers.html#regriddataplane
###

MODEL = QPF
OBTYP = QPE

REGRID_DATA_PLANE_VERIFY_GRID={INPUT_BASE}/met_test/data/sample_fcst/2005080700/wrfprs_ruc13_12
REGRID_DATA_PLANE_METHOD = BUDGET
```


METplus use cases

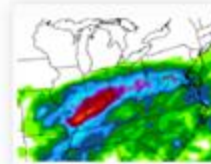
How many?! Over 100!

METplus use case applications / categories:

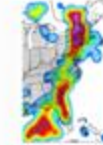
- ✓ CAM/CAE / Short-range
- ✓ Global / Medium-range
- ✓ S2S / SFS
- ✓ Tropical Cyclone
- ✓ Precipitation
- ✓ Fire
- ✓ Land surface
- ✓ Planetary boundary layer
- ✓ Ocean/Lake
- ✓ Marine and cryosphere
- ✓ Air quality
- ✓ Data assimilation
- ✓ Space weather
- ✓ Unstructured grids
- ✓ ...and growing!

7.2.10. Precipitation

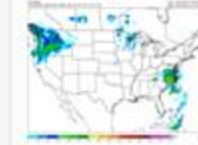
Any fields that can be defined as precipitation, including rain, snow, and other precipitation types



Grid-Stat: 24-hour QPF Use Case



MTD: Build Revision Series to Evaluate Forecast Consistency



Ensemble-Stat: WoFS



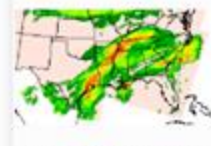
Grid-Stat: 6hr QPF in GEMPAK format



Grid-Stat: 6hr QPF in NetCDF format



PointStat: Compare community observed precipitation to model forecasts



Grid-Stat: 6hr PQPF Probability Verification



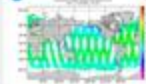
Point-Stat: Investigating Precipitation Types

7.2.7. Marine and Cryosphere

Data related to verification involving marine and cryosphere systems, which includes sea-ice



PointStat: read in satellite data and verify wind speeds or wave heights



GridStat: Python Embedding for sea surface salinity using level 3, 8 day mean obs



Grid-Stat and MODE: Sea Ice Validation



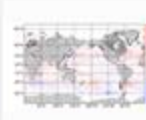
PointStat: read in buoy ASCII files to compare to model wave heights



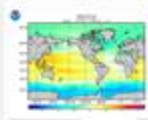
UserScript: Python Script to compute cable transport



PlotDataPlane: Python Embedding to read and process tripolar coordinate file



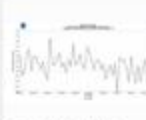
GridStat: Python Embedding to read and process SST



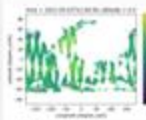
GridStat: Python Embedding to read and process sea surface heights



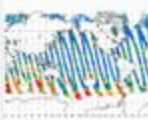
GridStat: Python Embedding to read and process ice cover



PointStat: Python embedding to read Argo netCDF files to verify ocean temperature forecast at 50 m depth



GridStat: Python Embedding for sea surface salinity using level 3, 1 day composite obs



PointStat: read in directory of ASCAT files over user-specified time

METplus use cases

What benefits do they have? Testing!

- Use cases contributed to the METplus repository are run as part of our automated testing workflow using GitHub Actions and Docker (*see John Halley Gotway's talk at 4:00 pm for more details!*)
- Changes to the METplus component repositories trigger all use cases to run to ensure changes do not break any current functionality or cause unintended changes
- Failures and differences in the output are flagged

METplus use cases

What benefits do they have? Reproducibility and sharing of configurations!

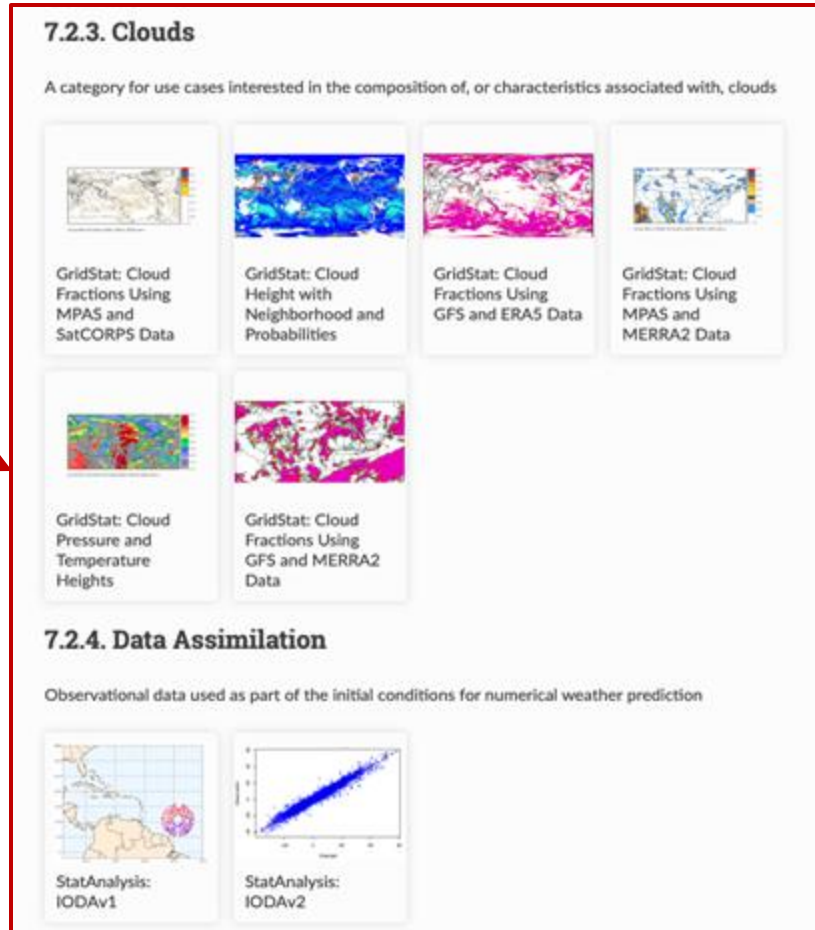
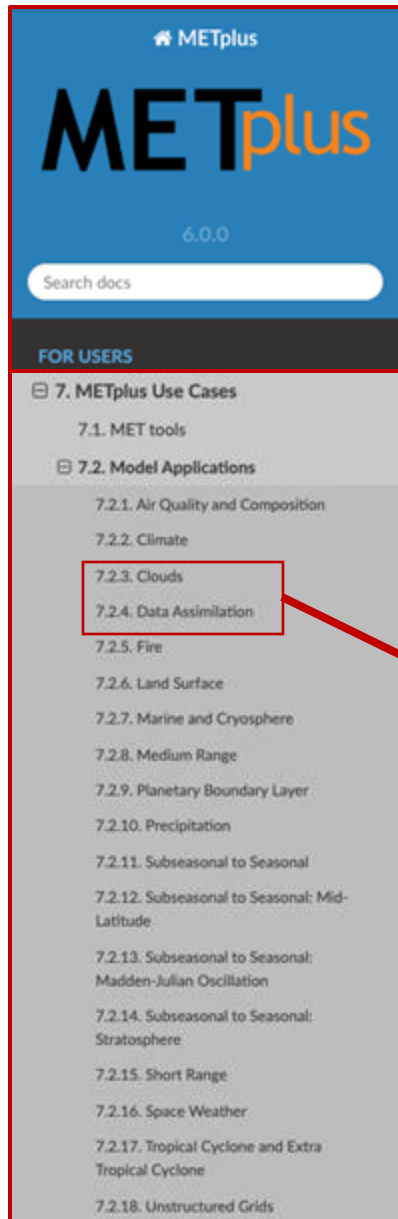
- ✓ Shared configuration files = consistent setups across users and systems
- ✓ Facilitates statistically rigorous, reproducible science
- ✓ Reduces configuration drift and human error
- ✓ Provides transparent, traceable workflows
- ✓ Speeds up METplus onboarding/training with ready-to-run examples
- ✓ Strengthens collaborations and community trust



Support, training, and resources

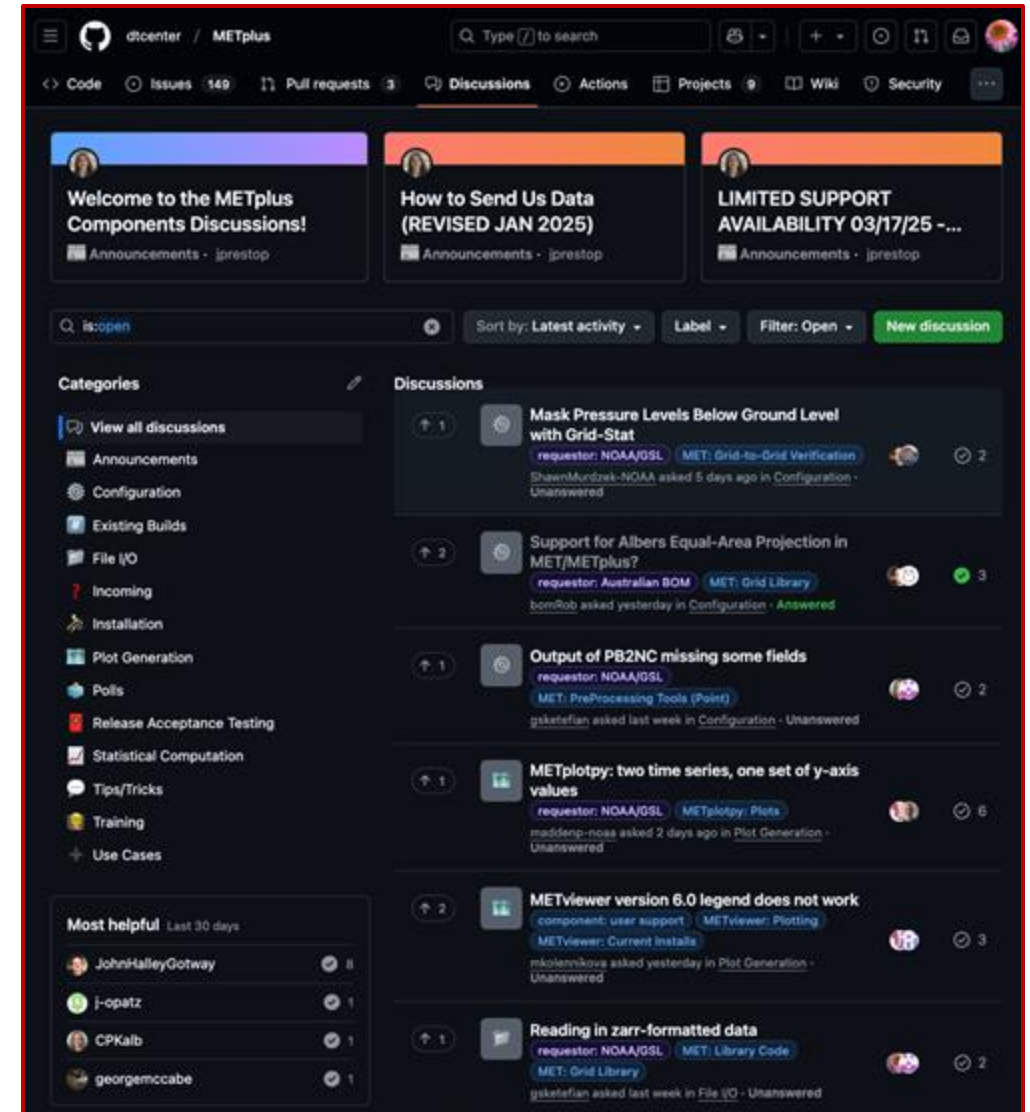
Users' Guides & Getting Help!

Access over 100 METplus use cases!



https://metplus.readthedocs.io/en/latest/Users_Guide/

Get help via METplus GitHub Discussions!



<https://github.com/dtcenter/METplus/discussions>

METplus resources

Recent/projected coordinated releases:

- ✓ Version 6.0 – December 2024
- ✓ Version 6.1 – July 2025
- 31 Version 6.2 – November 2025

Online Resources:

METplus [user support discussion forum](#)

METplus [website](#), [online tutorial](#)

METplus [repository](#), [documentation](#), [releases](#)



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